

Rishabh Kumar Pandey, Dr.

✉ rishabh@cuhk.edu.cn  rishabhp109@gmail.com
 <https://orcid.org/0000-0003-4074-9203>
 <https://scholar.google.com/citations?user=RRKER5cAAAAJ&hl=en>
 <https://www.linkedin.com/in/rishabh-kumar-pandey-7000b4a4/>
 <https://www.researchgate.net/profile/Rishabh-Pandey-27>
 +86 13265534517, +91 9795956882



Employment

- July 2025 –  **Post-doctoral fellow**, School of Science and Engineering (SSE), The Chinese University of Hong Kong, Shenzhen, China.
Supervisor: Dr Bingyu Cui
- Nov 2024 – May 2025  **Chemistry Researcher**
Turing Enterprises, Inc.
548 Market Street, PMB 18282
San Francisco, CA 94101

Education

- 2019 – 2024  **Ph.D., Indian Institute of Technology Patna**
Supervisor: Dr. Tammineni Rajagopala Rao, Associate Professor.
Thesis title: *Vibronic Coupling Theory and its Successful Applications in Photoelectron Spectra of Aluminium Clusters.*
- 2016 – 2018  **M.Sc. Chemistry, Mahatma Gandhi Post Graduate College, Gorakhpur**
Affiliated to Deen Dayal Upadhyay Gorakhpur University.
- 2012 – 2015  **B.Sc., Buddha Post Graduate College, Kushinagar.**
Affiliated to Deen Dayal Upadhyay Gorakhpur University.

Overview of projects

Currently, I am working on generalising the theories for ET rates under the strong light-matter coupling conditions. I am also experienced in quantum dynamics and its applications in reproducing the photo-electron spectrum through ab initio electronic structure calculations and quantum nuclear dynamics for different molecular systems. Solved the Hamiltonians for various polyatomic systems and examined the non-adiabatic effects present in them through potential energy curves and the nuclear dynamics studies. Also performed electronic structure calculations for different systems of interest.

Research Publications

Journal Articles

- 1** **R. K. Pandey** and B. Cui, “Vibronic Coupling Effects in the Photoelectron Spectra of LiB_6^- ,” *Physical Chemistry Chemical Physics*, vol. 28, pp. 15 942–15 953, 26 2026.  DOI: <https://doi.org/10.1039/D6CP01248E>.
- 2** **R. K. Pandey**, A. K. Kanakati, S. S. Tripathi, and T. R. Rao, “Vibronic interactions in the photoelectron spectra of CAI_3Ge^- : A theoretical study,” *The Journal of Physical Chemistry A*, vol. 129, pp. 14–27, 1 2025.  DOI: <https://doi.org/10.1021/acs.jpca.4c05354>.

- 3 S. S. Tripathi, K. Srikanth, **R. K. Pandey**, and T. R. Rao, "Theoretical Elucidation of Photoelectron Spectrum of Diphosphatriazolate Anion, $P_2N_3^-$," *The Journal of Physical Chemistry A*, vol. 129, pp. 7617–7629, 33 2025.  DOI: <https://doi.org/10.1021/acs.jpca.5c02662>.
- 4 A. Hassan, **R. K. Pandey**, A. Chakraborty, S. A. Wahed, T. R. Rao, and N. Das, "Green synthesis of an ionic porous organic polymer for efficient capture of environmentally toxic MnO_4^- and I_3^- from water," *Soft Matter*, vol. 20, pp. 7832–7842, 2024.  DOI: <https://doi.org/10.1039/D4SM00679H>.
- 5 **R. K. Pandey**, K. Srikanth, S. S. Tripathi, and T. R. Rao, "Resolving the Experimental Photoelectron Spectra of CAI_3Si^- ," *The Journal of Physical Chemistry A*, vol. 128, no. 2, pp. 355–369, 2024.  DOI: [10.1021/acs.jpca.3c06295](https://doi.org/10.1021/acs.jpca.3c06295).
- 6 A. Tak, K. Srikanth, **R. K. Pandey**, G. Guillon, P. Honvault, and T. R. Rao, "Quantum dynamical study on $^{15}N+^{14}N^{14}N$ isotope exchange reactions at high energies," *Chemical Physics Letters*, 2024.  DOI: <https://doi.org/10.1016/j.cplett.2024.141451>.
- 7 A. Chakraborty, S. Pandey, **R. K. Pandey**, *et al.*, "Self-assembled neutral and ionic [2+2] metallomacrocycles using a new flexible ditopic Pt(II)-based organometallic tecton bearing a pyrimidine motif: Facile syntheses and enhanced anticancer potency," *Applied Organometallic Chemistry*, vol. 37, no. 1, e6917, 2023.  DOI: <https://doi.org/10.1002/aoc.6917>.
- 8 **R. K. Pandey**, K. Srikanth, A. Tak, A. Kumar, and T. R. Rao, "A theoretical study of vibronic coupling in the photoelectron spectra of Al_6N^- ," *Physical Chemistry Chemical Physics*, vol. 25, pp. 12 990–13 003, 18 2023.  DOI: [10.1039/D3CP00836C](https://doi.org/10.1039/D3CP00836C).

Skills

Languages	 Strong reading, writing and speaking competencies for English. Quick learning ability.
Coding	 Fortran, Python, $\LaTeX$.
Softwares	 GAUSSIAN, MCTDH, MOLPRO, GAMESS, ORCA.
Web Dev	 HTML, Google Sites.
Misc.	 Academic research, Scientific temper, Teaching, $\LaTeX$ typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

- 2023  **ACS Best Poster Award** Theoretical Chemistry Symposium 2023 (TCS 2023), IIT Madras.
- 2019  Qualified **GATE 2019**

Conferences

- 2026  **International Conference on Photochemistry, Photo-physics and Photodynamics, IP³-2026, Shiv Nadar University, Delhi NCR.** Presented Poster.
- 2025  **The 26th International Conference on the Jahn-Teller Effect, JT 2025.** Presented Poster (Online).
- 2024  **Recent Trends in Chemical Science and Technology, RTCST-2024, IIT Patna.** Presented Poster.
- 2023  **Theoretical Chemistry Symposium 2023 (TCS 2023), IIT Madras.** Presented poster and won the 'ACS Best Poster Award'.
-  **An International conference on Structure and Dynamics: Spectroscopy and Scattering, SDSS-2023, IACS Kolkata.** Presented poster.
- 2022  **Current Trends in Theoretical Chemistry, CTTC-2022, BARC Mumbai.** Presented Poster.

Miscellaneous Experience (continued)

2020  **Current Trends in Theoretical Chemistry, CTTC-2020, BARC Mumbai.** Presented Poster (Online).

Workshops

2022  **Mumbai Workshop on Quantum Chemistry, MWQC 2022, IIT Bombay.** Give an oral presentation.

References

Dr. Tammineni Rajagopala Rao

Associate Professor
Department of Chemistry,
Indian Institute of Technology Patna,
Bihar, India 801106.
 rajgopal@iitp.ac.in

Dr. Sanjay Kumar

Professor
Department of Chemistry,
Indian Institute of Technology Madras.
Chennai, India 600036.
 sanjay@iitm.ac.in

Dr. Neeladri Das

Associate Professor
Department of Chemistry,
Indian Institute of Technology Patna.
Bihar, India 801106.
 neeladri@iitp.ac.in

Dr. Bingyu Cui

Assistant Professor
School of Science and Engineering,
The Chinese University of Hong Kong Shenzhen.
Guangdong, China 518172.
 bycui@cuhk.edu.cn